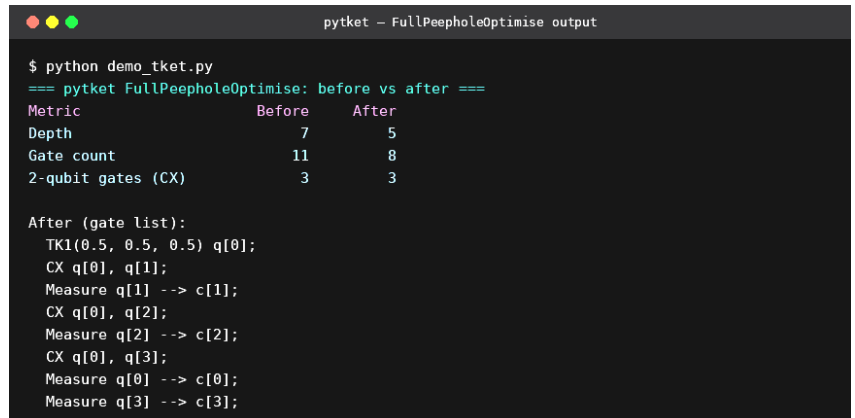


optimiser shows how much of that cost can be clawed back through algebraic simplification before or after routing.

All these projects aim at interoperability instead of settling on a single winner. Tket runs circuits in both Qiskit and Cirq. PennyLane offers plugins to interface with multiple backends. Cloud QPUs (IBM Quantum, Azure Quantum, Amazon Braket, etc) provide a choice of compilers as a pluggable layer, as opposed to a completely integrated and set stack. With improvements in hardware, focus in compiler research will likely shift to error correction compilation and resource estimation in conjunction with noise-aware, calibration driven optimisation.

Open source quantum compilation tools have advanced to the point where they can be considered serious competition. Qiskit, Cirq, tket and PennyLane, for instance, have a great deal of flexibility, each in their own ways. Most serious open source projects leverage multiple compilers. It is now essential to understand what each compiler optimises, and why, in order to get the best results from today's noisy quantum devices. **END** 



```

pytket - FullPeepholeOptimise output

$ python demo_tket.py
=== pytket FullPeepholeOptimise: before vs after ===
Metric                Before  After
Depth                 7      5
Gate count            11      8
2-qubit gates (CX)    3      3

After (gate list):
TK1(0.5, 0.5, 0.5) q[0];
CX q[0], q[1];
Measure q[1] --> c[1];
CX q[0], q[2];
Measure q[2] --> c[2];
CX q[0], q[3];
Measure q[0] --> c[0];
Measure q[3] --> c[3];

```

Figure 4: Actual console output: the optimiser collapsed the redundant H-H and zero-angle Rz into a single combined TK1 rotation, cutting depth from 7 to 5 and gate count from 11 to 8 while leaving the three CX gates untouched

References

- The Quantum Insider, 'State of Quantum Computing: Programming Languages', June 2026.
- Adnan Masood, 'The Quantum Platforms Briefing — Day 5: Open Source Ecosystem', Medium, October 2025.
- QuantumZeitgeist, 'Top Quantum Software Companies 2026: The Definitive Stack Guide', May 2026.
- PostQuantum.com, 'Quantum Programming: An In-Depth Introduction and Framework Comparison', September 2025.
- RankRed, '13 Quantum Programming Languages (& Tools) To Learn In 2026'.

By: Dr R. Anand

The author is a professor in the Department of Computer Science and Engineering at Easwari Engineering College, Chennai, Tamil Nadu.

■ ■ Continue from page...44

before measuring, we converted that invisible phase into a difference in outcomes that is not merely visible but absolutely certain. This is the same trick we used in the third article, when a second Hadamard gate exposed the hidden work of the Pauli-Z gate, and it has a name: quantum interference. Every quantum algorithm worth the name is, at bottom, an elaborate arrangement of this trick.

It is now time to conclude. In this article we executed an entangling circuit on a real 156-qubit superconducting processor and found 13 shots out of 1000 that had no business being there. Chasing those 13 counts took us through the whole zoo of errors that afflict present-day quantum hardware—relaxation, dephasing, gate errors, readout errors, crosstalk, leakage, material defects, and cosmic rays. We then met the four Bell states, proved what it means for a state to be entangled, discovered that measurement in the computational basis cannot tell all four apart, and repaired that failure with a Bell measurement.

That last experiment left us with a loose thread, and it is deliberately loose. Twice now we have made a hidden phase visible by applying an extra gate before measuring: once with a single qubit in the third article, and once with a Bell pair in this one. Both times the revealing gate was the Hadamard—the first gate we ever met, and the one we have used most often without examining it closely. In the next article we shall give the Hadamard operator the attention it deserves and then turn to the phenomenon it has quietly been exploiting all along: quantum interference. We shall see how the cancellation of unwanted amplitudes—not some fantasy of trying every answer at once—is the real engine of quantum speedup. Entanglement, it turns out, was only half the story. **END** 

By: Dr Deepu Benson

The author is a free software enthusiast with a keen interest in theoretical computer science. He received his PhD from IIT Palakkad and completed his postdoctoral research at IIT Gandhinagar. He maintains a technical blog.